

Article

Stability and Optimization of a Vector Thrust-Controlled Tail-Sitter UAV Based on Flight Test

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Highlights

What are the main findings?

- A nonlinear dynamic model characterizing the coupling between longitudinal aerodynamics and vector propulsion was established and validated against flight tests.
- A hybrid optimization strategy utilizing pattern search and SQP was employed to optimize control parameters, resulting in significantly enhanced controller performance.

What are the implications of the main findings?

- The validated model provides a systematic framework for the dynamic modeling of vector-thrust UAVs.
- The proposed optimization method offers an effective solution for the optimization of UAV control parameters.

Abstract

Stability plays essential roles for Vertical Take-Off and Landing (VTOL) vehicles. This paper investigates the stability characteristics of a novel tail-sitter VTOL vehicle employing vector thrust control, specifically focusing on nonlinear modeling and parameter optimization. Firstly, the tail-sitter VTOL which employs vector thrust controlling principles, is designed, and manufactured using 3D printing and carbon-fiber reinforced techniques, with a customized flight controller implemented on the PX4 architecture. To address the nonlinear dynamic characteristics introduced by the vector thrust mechanism, a nonlinear dynamic model for cruise flight is established based on an offline database and validated against cruise flight test data. Flight tests show that the vector-thrust-based pitch control provides rapid response and accurate tracking during cruise flight. Furthermore, based on the validated model, a hybrid optimization strategy combining pattern search and sequential quadratic programming (SQP) is used to tune the cascaded control parameters. Simulation results demonstrate that the optimized controller reduces the rise time from 6.8 s to 0.2 s and the settling time from 10.1 s to 0.9 s under the tested cruise-condition step response, indicating a marked improvement in dynamic response performance. This study provides a practical framework for cruise-flight modeling, pitch-stability analysis, and control-parameter optimization of vector-thrust tail-sitter UAVs.

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Keywords: tail-sitter UAV; vector thrust; nonlinear modeling; parameter optimization; flight test

1. Introduction

Unmanned Aerial Vehicles (UAVs) have witnessed significant adoption across both military and civilian domains, playing pivotal roles in aerial surveillance, logistical delivery, and environmental monitoring [1]. Conventionally, UAVs are categorized into fixed-wing and rotary-wing configurations. While fixed-wing aircraft are renowned for their high cruising speed and aerodynamic efficiency, their dependence on runways or launch catapults limits deployment in complex terrains [2]. Conversely, rotary-wing UAVs offer VTOL and hovering capabilities but suffer from limited endurance and lower flight speeds due to high energy consumption. To bridge the gap between these distinct platforms, hybrid VTOL UAVs have emerged as a prominent research focus, aiming to integrate the endurance of fixed-wing aircraft with the operational flexibility of multi-rotors [3].

Among hybrid configurations, the tail-sitter UAV represents a highly efficient solution. By utilizing a unified propulsion system for both vertical and horizontal flight, it eliminates the dead weight and parasitic drag associated with the redundant mechanisms found in tilt-rotor or composite-wing designs [4]. To further enhance control authority, vector thrust technology is increasingly adopted in modern tail-sitter designs [5]. This mechanism generates control moments by directly deflecting the thrust vector, ensuring robust maneuverability and stability even at low airspeeds where aerodynamic surfaces are ineffective [6].

Despite these aerodynamic advantages, the flight control of vector-thrust tail-sitters presents substantial engineering challenges. The vehicle must operate across a wide flight envelope involving drastic changes in the angle of attack. During the transition and cruise phases, the dynamic behavior is governed by a complex interplay between nonlinear aerodynamic forces and vector thrust moments [7]. This strong coupling creates a nonlinear dynamic environment that is difficult to describe accurately using simplified linear models. Consequently, establishing a high-fidelity nonlinear dynamic model that explicitly accounts for vector thrust characteristics is a prerequisite for ensuring flight stability [8].

Existing studies have investigated VTOL and tail-sitter UAVs from different perspectives. Li et al. developed an MPC controller for a tail-sitter UAV in hover flight based on a six-degree-of-freedom nonlinear model obtained from wind-tunnel aerodynamic data, and validated the controller through HIL simulation and indoor flight experiments [9]. Their study demonstrated the effectiveness of model-based hover-flight control, but did not address cruise-phase longitudinal modeling and controller-parameter optimization. Durán-Delfín et al. presented a mathematical model and a passivity-based controller for a convertible fixed-wing VTOL, focusing on transition flight and numerical simulation validation [10]. In contrast, the present study considers a vector-thrust tail-sitter UAV whose cruise-phase longitudinal control is governed primarily by integrated vector thrust rather than by conventional elevator-based aerodynamic control. Therefore, the main objective of this work is not a broad benchmark of advanced controllers, but the modeling, flight-test-supported validation, and constrained parameter tuning of this specific cruise-phase longitudinal control mechanism.

Currently, the proportional-integral-derivative (PID) controller remains the standard industrial solution for stabilizing such platforms due to its simplicity and model independence. However, tuning PID parameters for a highly coupled, nonlinear vector-thrust system is a non-trivial task [11]. Traditional manual tuning relies heavily on trial-and-error flight tests, which are not only time-consuming and inefficient but also pose significant safety risks

to the prototype. While various automated tuning methods exist, they face distinct limitations: gradient-based algorithms (e.g., Gradient Descent) are prone to getting trapped in local minima due to the non-convex nature of the objective function, whereas global search algorithms (e.g., Genetic Algorithms or Particle Swarm Optimization) often suffer from slow convergence and excessive computational costs [12]. In the area of controller tuning, Su et al. showed that hybrid optimization combining gradient-based methods and pattern search can be effective for VTOL controller design when supported by system identification and flight testing [13]. In contrast, Bazzocchi et al. proposed an offline autopilot tuning framework based on system identification, control matching, and genetic algorithms, which illustrates the practicality of pure global-search optimization for UAV controller tuning [14]. These studies suggest that pure global-search methods are robust for broad parameter exploration, whereas hybrid strategies are advantageous when both computational efficiency and local refinement are required. Motivated by this trade-off, the present study adopts a two-stage strategy in which pattern search provides coarse exploration and SQP performs local refinement for the nonlinear constrained tuning problem considered here.

To address these challenges, this paper presents a systematic framework for the modeling and parameter optimization of the ETS30, a custom-designed vector-thrust tail-sitter UAV. The main contributions of this study are as follows: (1) A nonlinear dynamic model characterizing the longitudinal aerodynamic and vector propulsion coupling effects was established. By integrating the OpenVSP aerodynamic database, the model's fidelity was rigorously verified against flight test data. (2) A two-stage hybrid optimization strategy combining Pattern Search and Sequential Quadratic Programming (SQP) was utilized. This method effectively balances global exploration with local refinement to identify the optimal control parameters within a constrained feasible region.

2. The Vector-Controlled Tail-Sitter UAV

2.1. Platform Design and Configuration

The ETS30 is a custom-developed tail-sitter VTOL unmanned aerial vehicle developed by Southern University of Science and Technology (as shown in Figure 1). The airframe is constructed using a hybrid manufacturing approach, utilizing carbon fiber reinforced polymers for the main load-bearing structures and 3D-printed components for aerodynamic fairings, achieving an optimal balance between structural rigidity and lightweight design.

The propulsion system features a tri-rotor configuration tailored for multi-stage flight. Two auxiliary propellers (vector thrust 1 and 2) are mounted on the trailing edge of the main wing, while a primary propulsion unit (vector thrust 3) is located at the tail. During the vertical takeoff and landing phases, all three propellers operate simultaneously to provide sufficient lift. Upon transitioning to the level-flight cruise mode, the two wing-mounted propellers are deactivated and folded to minimize aerodynamic drag, leaving only the tail propeller to provide forward thrust. The present study therefore focuses on the cruise phase, in which the wing-mounted auxiliary propellers are deactivated and the longitudinal dynamics are governed primarily by the tail vectoring unit. This operating condition allows the pitch-stability mechanism introduced by vector thrust to be isolated and directly compared with flight-test data.

A distinct feature of the ETS30 is its exclusive reliance on vector thrust for longitudinal control in cruise mode, eliminating the need for conventional aerodynamic control surfaces such as elevators. This design significantly simplifies the mechanical complexity of the wing structure while enhancing control authority through direct force manipulation.

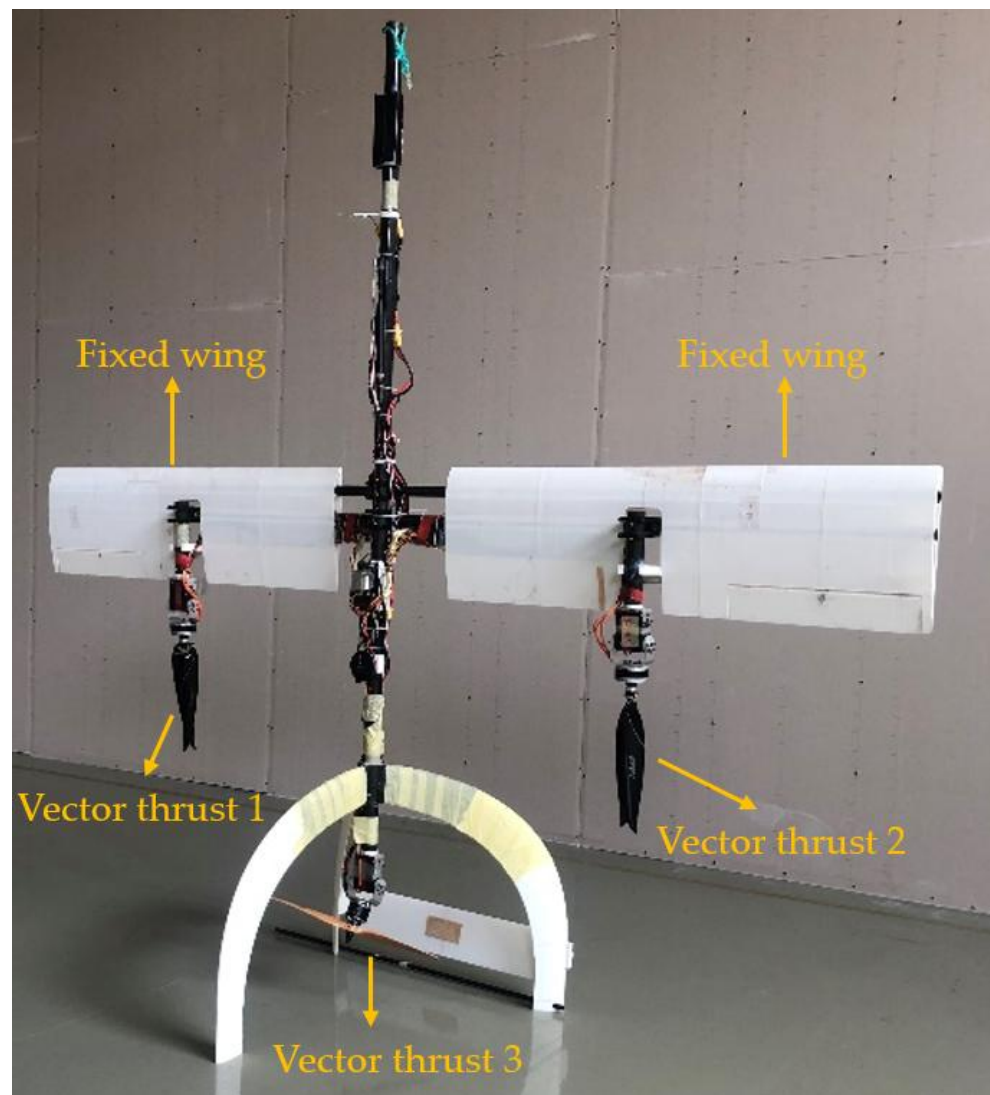


Figure 1. The ETS30 vector-thrust tail-sitter UAV platform.

2.2. Propulsion and Vectoring Mechanism

The propulsion system of the ETS30 consists of three motor-propeller groups designed to operate in distinct modes corresponding to different flight phases. In the cruise mode, the primary thrust is provided exclusively by the independent tail propulsion unit. This unit integrates thrust vectoring capability and serves as the core actuator for longitudinal control without relying on aerodynamic surfaces.

The vectoring mechanism employs a single-axis tilt design where a servo drives the tail motor mount to deflect around the lateral axis of the airframe. When the servo is actuated, the direction of the thrust vector changes, thereby generating a force component perpendicular to the body axis. Due to the significant lever arm between the tail motor installation point and the center of gravity (CG), this force component produces a direct and highly efficient pitching moment. This mechanism based on direct force control ensures that the aircraft maintains rapid and precise attitude response capabilities throughout the cruise envelope.

The onboard data acquisition system incorporates various sensors, including accelerometers, GPS, gyroscopes, and barometers, all of which are systematically summarized in Table 1. The basic physical parameters of the ETS30 platform adopted in the cruise longitudinal model are summarized in Table 2.

Table 1. The specifications of the onboard sensors.

SN	Description	Model	Accuracy
1	GPS	NEOV3	±0.5–2m
2	Barometer	MS5611	±1.5 mbar
3	Accelerometer and gyroscope	ICM-20689	±2%, ±2%
		ICM-20602	±1%, ±1%
		BMI055	0.98 mg, ±0.004 deg/s
4	Magnetometer	IST8310	±0.3 uT

Table 2. Basic physical parameters of ETS30.

Parameter	Value
m	7.7 kg
S	0.4325 m ²
c	0.25 m
I_{yy}	1.5 kg·m ²
l_t	0.8 m
CG location	−0.12/c
Maximum vector deflection, $\delta_{vec,max}$	45 deg

3. Nonlinear Dynamic Modeling and Control Architecture

3.1. Modeling Assumptions and Coordinate Systems

To characterize the longitudinal dynamic behavior of the ETS30 UAV during the cruise phase, a nonlinear three-degree-of-freedom point-mass model is established. The modeling framework relies on the following key assumptions: (1) Rigid Body Assumption: The aircraft is treated as a rigid body, and its mass properties are assumed to be invariant throughout the simulation horizon. (2) Longitudinal Decoupling: The motion is restricted strictly to the vertical plane (longitudinal motion), neglecting any coupling effects with lateral-directional dynamics. These assumptions are considered appropriate for the baseline cruise condition addressed in this study, where sideslip is small and roll–yaw coupling is not dominant in the measured longitudinal response. The force analysis of the ETS30 is illustrated in Figure 2.

Accordingly, the system state vector is defined as $x = [V, \alpha, q, \theta, h]^T$, representing air-speed, angle of attack, pitch rate, pitch angle, and altitude, respectively. The equations of motion are formulated in the Wind Axes to facilitate the direct integration of the aerodynamic database. In Figure 2, the angle of attack α is defined as the angle between the body axis and the velocity vector, while the flight-path angle γ denotes the inclination of the velocity vector with respect to the inertial horizontal direction. For the baseline cruise-validation condition considered in this study, the thrust-vector deflection remains modest; accordingly, its direct contribution to the translational force balance is treated as a secondary correction in the present engineering model, while its pitching-moment contribution is retained explicitly. The present formulation is restricted to the cruise-phase longitudinal channel of the ETS30, while hover and transition maneuvers, which involve additional multi-propeller coupling and mode-switching effects, are outside the scope of the current model. The main symbols used in the model are summarized in Appendix A.

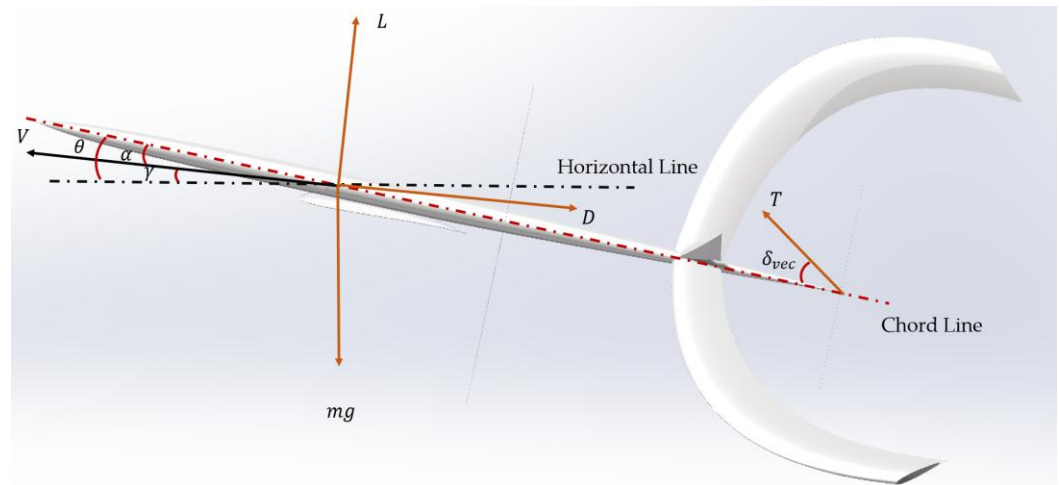


Figure 2. Force analysis diagram.

3.2. Aerodynamic and Propulsion Database

The aerodynamic forces and propulsive performance are derived from offline look-up tables rather than simplified linear derivatives. Cubic spline interpolation is employed for real-time data retrieval during the simulation. The present aerodynamic database is intended for engineering-level cruise modeling and has not been independently calibrated against wind-tunnel, CFD, or flight-identification data.

The aerodynamic coefficients are sourced from an aerodynamic database generated via OpenVSP. The lift coefficient C_L and drag coefficient C_D , and total pitching moment coefficient C_m are modeled as follows:

$$C_L(\alpha) = C_{L,interp}(\alpha) \quad (1)$$

$$C_D(\alpha) = C_{Di}(\alpha) + C_{D0} \quad (2)$$

$$C_m(\alpha) = C_{m,base}(\alpha) + C_{mq} \frac{qc}{2V} - \frac{T \sin(\delta_{vec}) l_t}{QSc} + C_{m0} \quad (3)$$

Lift coefficient (C_L): Directly interpolated from the OpenVSP database. Drag coefficient (C_D): Modeled using a component build-up method. It consists of the induced drag (C_{Di}) retrieved from the OpenVSP database and a constant zero-lift parasitic drag coefficient (C_{D0}). Static pitching moment coefficient ($C_{m,base}$): The raw moment coefficient from the database is referenced to the wing leading edge. The model mathematically corrects this by transferring the moment to the actual Center of Gravity (CG), accounting for the lever arm between the reference point and the CG.

The propulsion system is modeled using experimental performance data tables. The thrust is determined by interpolating the dataset, which characterizes propeller performance as a function of freestream velocity and rotational speed. This allows the variation in thrust with freestream velocity and rotational speed to be represented in the simulation.

The total pitching moment coefficient (C_m) integrates the static aerodynamics with dynamic damping and vector thrust control effects. Specifically, Equation (3) is written in coefficient form as the sum of the static aerodynamic pitching moment, the pitch-rate damping contribution, the thrust-vectoring control contribution, and the intrinsic trim term. The static pitching moment is referenced to the actual center of gravity after transferring the raw OpenVSP moment coefficient from the wing leading-edge reference. The dynamic damping term is expressed in the standard nondimensional form $\frac{qc}{2V}$, where q is the pitch rate, c is the mean aerodynamic chord, and V is the instantaneous airspeed. Dynamic damping ($C_{mq} \frac{qc}{2V}$) represents the aerodynamic damping moment generated by

the pitch rate. Vector thrust control $(-\frac{T\sin(\delta_{vec})l_t}{QSc})$ quantifies the control moment generated by the thrust vectoring mechanism. Here, T is the interpolated thrust, δ_{vec} is the vector deflection angle, and l_t is the lever arm from the tail motor to the CG. This term is normalized by the dynamic pressure, wing area, and mean aerodynamic chord. Here, the thrust-vectoring moment term is converted into an equivalent pitching-moment coefficient through normalization by QSc , where $Q = \frac{1}{2}\rho V^2$ is the dynamic pressure, ρ is the air density, and V is the instantaneous airspeed during the simulation. Intrinsic trim (C_{m0}) represents the inherent residual moment coefficient of the airframe. Accordingly, the interpolated aerodynamic coefficients should be interpreted as engineering-level cruise-condition inputs rather than independently verified aerodynamic truth data. Although cubic spline interpolation may introduce interpolation and smoothing uncertainty, this uncertainty is considered acceptable for the present study because the model is used for cruise longitudinal response analysis and controller tuning, and its practical adequacy is subsequently supported by the flight-test comparison in Section 5.

3.3. Longitudinal Equations of Motion

Applying Newton's Second Law and the theorem of angular momentum in the Wind Axes, the nonlinear longitudinal equations of motion for the ETS30 in the cruise phase are derived. These equations characterize the translational and rotational dynamics under the combined influence of aerodynamics, gravity, and vector thrust. In this formulation, the translational equations are expressed in the Wind Axes, whereas the rotational dynamics are driven by the total pitching moment M , whose coefficient form is defined in Equation (3).

$$m\dot{V} = T_c \cos(\alpha) - D - mg \sin(\gamma) \quad (4)$$

$$mV(\dot{\theta} - \dot{\alpha}) = L + T_c \sin(\alpha) - mg \cos(\gamma) \quad (5)$$

$$I_{yy}\ddot{\theta} = M \quad (6)$$

$$\dot{\theta} = q \quad (7)$$

$$\dot{h} = V \sin(\gamma) \quad (8)$$

where $\gamma = \theta - \alpha$ denotes the flight path angle and $T_c = T \cos(\delta_{vec})$ represents the effective propulsive component of the thrust along the body axis. Accordingly, the terms $T_c \cos(\alpha)$ and $T_c \sin(\alpha)$ in Equations (4) and (5) denote the projections of this effective body-axis thrust along and normal to the wind axis, respectively. Here, δ_{vec} denotes the deflection angle of the tail motor relative to the body axis, i.e., the vector-thrust actuator angle introduced in Section 2.2. Therefore, vector-thrust deflection influences the longitudinal dynamics in two ways: it changes the thrust contribution to the translational force balance through force decomposition, and it simultaneously generates a direct pitching moment through the tail moment arm. Equation (4) governs the tangential acceleration. The motive force for acceleration is derived from the projection of thrust onto the wind axis, while resistance arises from aerodynamic drag D and the gravitational component along the flight path $mg \sin(\gamma)$. Equation (5) describes the force balance perpendicular to the velocity vector, determining the rate of turn. It indicates that the centripetal force is provided by the aerodynamic lift (L), the gravitational component normal to the path ($mg \cos(\gamma)$), and crucially, the lift component contributed by the thrust ($T_c \sin(\alpha)$). This term highlights that for a tail-sitter, the propulsion system actively contributes to lift generation, especially at high angles of attack. For the baseline cruise-validation condition considered in this study, the vector-deflection amplitude remains modest. Accordingly,

the direct normal-force contribution associated with the vectored-thrust side component is treated as a secondary correction in the translational dynamics, whereas the corresponding pitching-moment contribution is retained explicitly in Equation (3) because it directly determines the longitudinal control authority through the tail moment arm. Equation (6) describes the rotational dynamics around the body Y-axis, driven by the total pitching moment M . Equation (7) defines the kinematic integral relationship between the pitch angle and pitch rate. Equation (8) indicates that the rate of altitude change is determined by the airspeed and the flight path angle.

The derived nonlinear equations of motion serve as the computational core for the ETS30 simulation environment. Ensuring the fidelity of this model is a prerequisite for the effective optimization of control parameters presented in Section 4. Consequently, the model has been validated against flight test data to verify its dynamic accuracy. A comprehensive comparative analysis between the simulation results and empirical telemetry is detailed in Section 5.

3.4. Control Architecture

To address the challenges of nonlinearity, actuator lag, and potential integral saturation caused by aggressive maneuvers during the cruise phase, a Cascaded PID Control Architecture with Feedforward Compensation and Anti-windup mechanisms is designed, as illustrated in Figure 3.

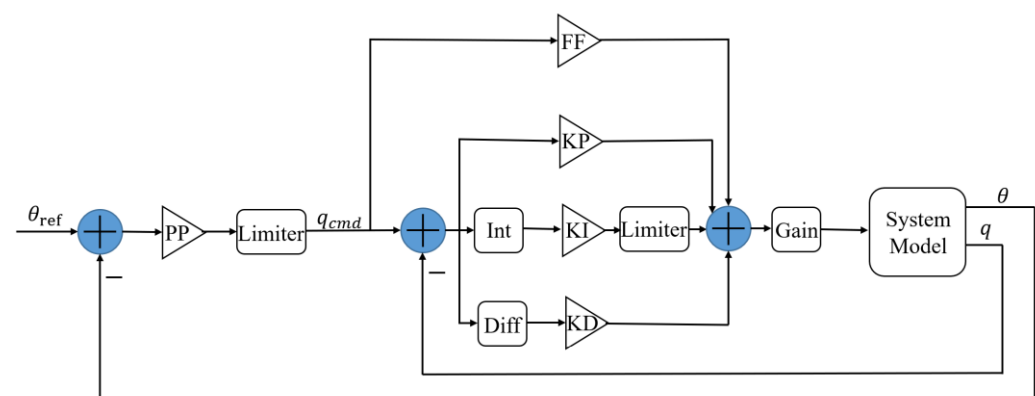


Figure 3. Longitudinal attitude control.

The choice of cascaded P-PID architecture is motivated by both practical implementation and modeling considerations. First, this control structure is consistent with the onboard flight-control architecture used in the ETS30 flight tests, which makes the simulation, validation, and parameter-tuning process directly comparable to the actual platform. Second, for the cruise longitudinal channel considered here, a cascaded PID-based controller provides sufficient control accuracy and disturbance rejection with low implementation complexity. Third, the nonlinear model developed in this study is intended as an engineering-oriented cruise model with several simplifying assumptions, rather than as a full high-fidelity representation of the entire flight envelope. Under this context, a cascaded P-PID controller is more appropriate than strongly model-dependent advanced controllers, because it remains effective even in the presence of practical modeling errors and unmodeled dynamics.

Advanced approaches such as predictive-control-related methods and passivity-based control have also been reported in the literature for other control problems and UAV configurations [15,16]. Specifically, ref. [15] discusses feed-forward design in predictive control for setpoint tracking and constraint handling, while ref. [16] presents passivity-based control laws for an unmanned powered parachute aircraft and emphasizes

robustness under mass variation and wind disturbance. However, the main objective of the present study is not a systematic comparison among different controller types, but the modeling, flight-test-supported validation, and constrained parameter tuning of the cruise-phase pitch motion of a vector-thrust tail-sitter UAV. Under this objective, the cascaded P-PID architecture remains the most appropriate choice because it is consistent with the actual ETS30 flight-control architecture and enables direct comparison between simulation and flight-test data.

The architecture comprises an outer attitude controller and an inner angular rate controller, configured in a proportional-proportional-integral-derivative (P-PID) structure. As shown in Figure 3, the control system adopts a dual closed-loop topology. The outer loop (attitude loop) is responsible for converting the pitch-angle error into a desired body-axis pitch-rate command to ensure stable attitude tracking. The inner loop (rate loop) is responsible for rapidly responding to rate commands, suppressing aerodynamic disturbances, and compensating for servo dynamic delays via a feedforward channel.

Outer loop strategy: The input to the outer loop is the deviation between the reference pitch angle and the measured value. A Proportional (P) control strategy is employed to map the angular error linearly to a pitch rate request. To prevent excessive angular rates that could exceed structural limits or the aerodynamic envelope during large deviations, a hard saturation logic (Limiter) is integrated into the control law. When the computed desired pitch rate exceeds the preset physical maximum, the system forcibly clamps the command output, ensuring flight safety. **Inner loop strategy:** The inner loop control strategy consists of parallel feedback and feedforward paths. **Feedback path:** Utilizes a PID algorithm to regulate the pitch rate error. Specifically, the proportional term provides the baseline response, the Integral term eliminates steady-state error, and the derivative term suppresses oscillations and enhances system damping. **Feedforward path:** To address the physical response lag of the thrust vectoring mechanism, a feedforward control is introduced. This channel directly maps the desired rate command to the control output proportionally. In the inner loop, the PID branch acts on the pitch-rate error, while the feedforward branch acts directly on the desired rate command, and the two outputs are summed to form the final actuator command. This mechanism improves the transient response speed by providing additional control action directly from the command signal.

4. The Optimization Methodology

4.1. Objective Function Formulation

To achieve optimal longitudinal flight performance for the ETS30 in cruise mode, precise tuning of the control parameters is essential. However, given the multi-parametric nature of the cascaded architecture, traditional manual tuning is not only time-consuming but also does not provide a systematic route to a high-quality feasible solution. To address this, a hybrid optimization strategy is employed to identify the optimal parameter set that maximizes longitudinal flight performance within the fixed control architecture.

For this problem, the controller gain vector to be optimized is defined as K :

$$K = (PP, KP, KI, KD, FF)$$

The primary objective is to minimize the tracking error e between the reference command and the simulation output while simultaneously suppressing high-frequency control oscillations. To quantify this performance, a composite cost function $J(K)$ is designed:

$$J(K) = \int_{t_1}^{t_2} w e(t)^2 dt + \int_{t_1}^{t_2} \lambda u(t)^2 dt + \eta \sum_i [\max(0, P_i)]^2 \quad (9)$$

This function incorporates three weighted terms: the integral of the square error, a control energy penalty, and a soft constraint penalty. The first term calculates the integral

of the square error (ISE) to strictly penalize any deviation from the target pitch angle, ensuring tracking precision:

$$e(t) = \frac{\theta_{sim}(t) - \theta_{base}(t)}{\theta_{step}} \quad (10)$$

where $\theta_{sim}(t)$ is the simulated pitch angle, and $\theta_{base}(t)$ is the reference pitch angle command, and θ_{step} is the step angle. By integrating the normalized error, the optimizer is driven to minimize the cumulative deviation throughout the entire flight process, ensuring that the UAV closely follows the target trajectory. In this item, w is a weight factor with a default value of 1. The second term accounts for the control energy cost, introduced to prevent high-frequency oscillations and reduce mechanical wear on the actuators. Here, $u(t)$ represents the normalized control input signal generated by the controller, ranging from -1 to 1 . The weight factor λ is set to 0.5 , which is the minimum value used to avoid high-frequency oscillations in the optimized response. Smaller values were found to favor more aggressive control commands and noticeable high-frequency oscillation in the actuator input, whereas excessively large values unnecessarily slowed the response. This term penalizes excessive or aggressive control actions by integrating the square of the control input over time. Minimizing this term ensures that the optimized control law generates smooth and efficient control commands, avoiding actuator saturation and extending the lifespan of the vectoring mechanism. Finally, the third term incorporates soft constraint penalties to enforce strict compliance with design specifications (detailed in Section 4.2). If a dynamic metric violates its constraint, a heavy penalty is added to the cost, forcing the optimizer to steer the solution back into the feasible region. Specifically, this penalty is defined as the product of the constraint violation P and a large weighting factor η (set to 10^5), so that any infeasible solution incurs a dominant cost and is rejected during the optimization process. The purpose of this setting is not to claim a uniquely optimal penalty value, but to ensure that any violation of the prescribed dynamic constraints immediately dominates the total cost, so that the optimizer is guided to remain in the feasible region before seeking further improvement in the tracking objective.

4.2. Dynamic Response Constraints

For the optimization problem, in addition to the objective function, constraints are required. Figure 4 illustrates a typical dynamic response, and the response characteristics define the feasible region for the objective solution. In the present study, the rise time t_r is defined as the time required for the normalized response to increase from 10% to 90% of the command step, the settling time t_s is evaluated using a $\pm 5\%$ error band around the target value, and the overshoot is defined as the peak exceedance relative to the command amplitude. These dynamic metrics are extracted automatically from the simulated step response during each optimization run and are used to determine whether a candidate parameter set satisfies the prescribed response constraints. These constraint bounds are used to avoid excessively slow response, large overshoot, and oscillatory behavior in the cruise longitudinal control tuning task. They were not introduced as arbitrary numerical limits, but as practical time-domain requirements associated with acceptable short-period pitch-response quality in cruise flight. As discussed in flight-dynamics literature, short-period flying qualities are closely related to response rapidity and damping characteristics [17]. Under this consideration, the imposed bounds on rise time, settling time, and overshoot are used here as engineering surrogates to exclude sluggish, weakly damped, or overly aggressive pitch responses during the optimization process, while remaining consistent with the response characteristics expected for the ETS30 platform in cruise longitudinal control.

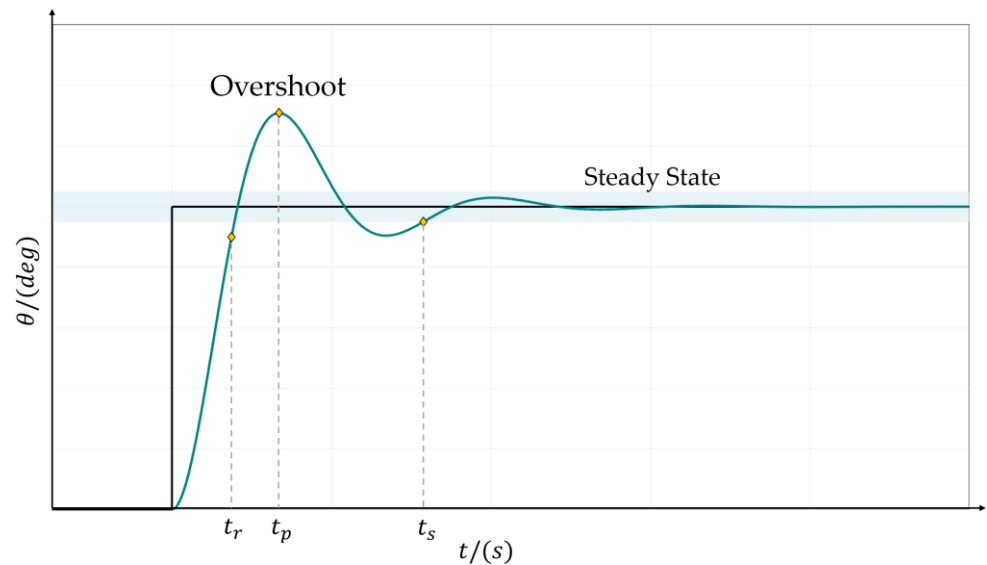


Figure 4. Dynamic response.

The constraints are detailed below. The optimization problem is thus defined as finding the parameter set K that minimizes the objective function value under these constraints:

$$J(K)$$

subject to

$$\begin{cases} t_r \leq 0.5 \\ t_s \leq 1.5 \\ M_p \leq 0.1 \\ |e_{ss}| \leq 0.05 \\ K \in [lb, ub] \end{cases}$$

where lb and ub represent the lower and upper bounds of the design parameters, respectively; M_p denotes the maximum overshoot; and e_{ss} represents the steady state error. Table 3 shows the bounds of controller gains. These gain bounds were selected around the baseline controller values to allow sufficient tuning range while avoiding excessively aggressive parameter combinations.

Table 3. Bounds of controller gains.

Parameter	Baseline	Min	Max
PP	2	1	10
KP	0.35	0.1	5
KI	0.2	0	2
KD	0	0	0.5
FF	0.15	0	5

4.3. Hybrid Optimization Method

The objective function landscape for the nonlinear tail-sitter UAV is often non-convex and contains multiple local minima. A single gradient-based algorithm relies heavily on the initial guess and is prone to being trapped in local optima, while a pure global search algorithm is computationally expensive. To balance computational efficiency and optimization accuracy, this paper proposes a two-stage hybrid optimization strategy that combines pattern search and sequential quadratic programming, as illustrated in Figure 5.

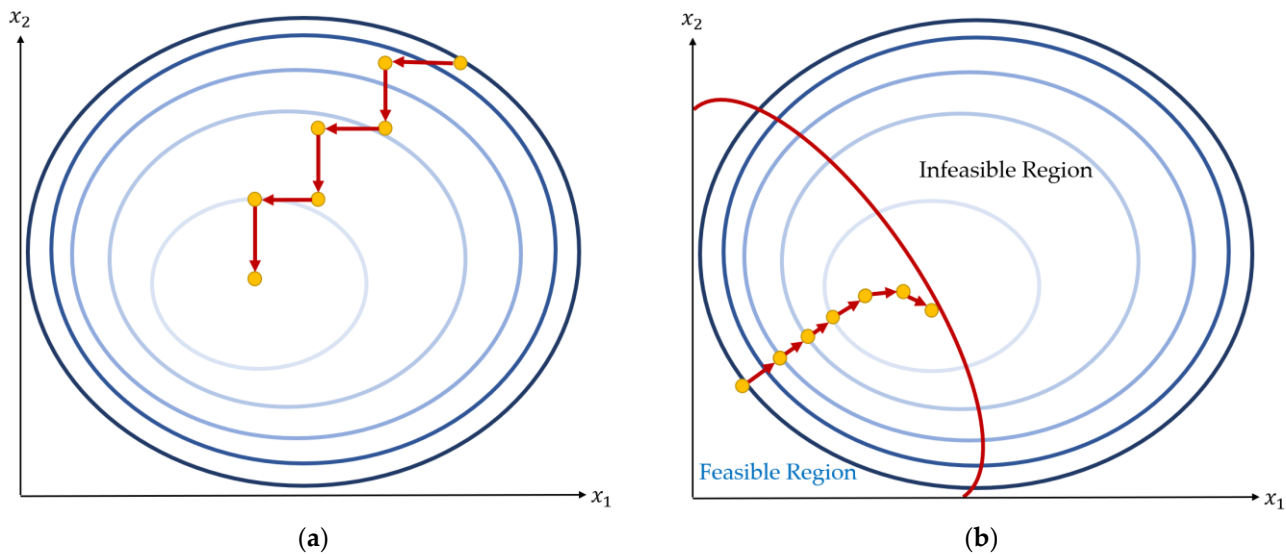


Figure 5. Schematic illustration of the optimization algorithms. The blue circles represent contour lines, and the red arrows indicate conceptual iterative paths. (a) Pattern search. The iteration path illustrates the direct search process across the objective function contours without gradient information; (b) SQP. The SQP algorithm efficiently navigates the feasible region boundary to locate the optimal solution where the objective gradient is counterbalanced by the constraint gradient.

The optimization process begins with a coarse global search using the pattern search algorithm. As a direct search method, it operates without gradient information and explores the parameter space by polling a mesh of points around the current iterate. The primary function of this stage is to robustly navigate the global search space to escape poor local solutions and locate a promising region for subsequent refinement.

Upon convergence of the first stage, the process transitions to the second stage of local refinement using sequential quadratic programming. This efficient gradient-based method approximates the objective function with a quadratic subproblem at each step. By employing the output of the first stage as a high-quality initial guess, the SQP algorithm utilizes gradient information to precisely refine the parameters, enabling efficient local refinement of the controller parameters.

5. Results and Discussion

To comprehensively evaluate the effectiveness of the proposed full vector-thrust control strategy and the hybrid optimization methodology, this section presents a systematic analysis of both simulation and experimental results. First, the fidelity of the established nonlinear dynamic model is verified against flight test data. Subsequently, a comparative analysis was performed to evaluate the control performance of the baseline and optimized controllers.

5.1. Model Verification Based on Flight Tests

Before performing parameter optimization, flight tests were conducted to validate the dynamic characteristics of the ETS30 prototype and the fidelity of the established nonlinear simulation environment. The corresponding flight-test video is provided in the Supplementary Materials as Video S1.

The flight data (black lines in Figure 6) demonstrates that the aircraft exhibits excellent command tracking capability even under baseline control conditions. The pitch attitude follows the reference signals promptly with no significant divergence or instability.

This confirms that the vector thrust mechanism provides sufficient control authority to maintain stable cruise flight.

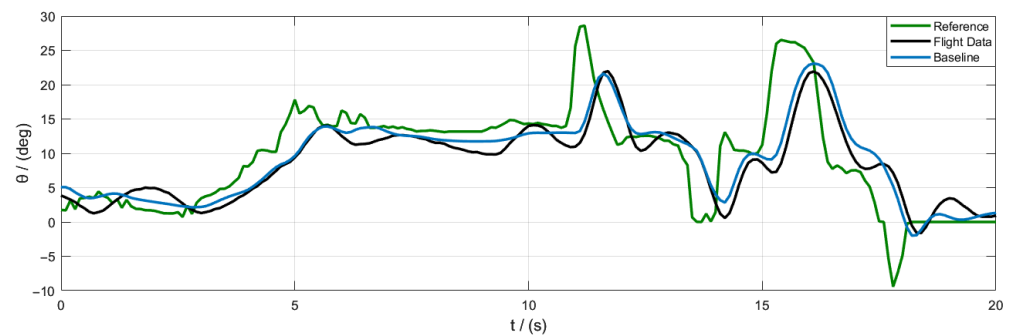


Figure 6. Comparison of flight telemetry data and baseline simulation response for model fidelity verification.

To verify the accuracy of the mathematical model established in Section 3, the recorded pitch-angle reference in the onboard flight-control logs was used as the command input to the nonlinear simulation model. The pitch attitude and rate used for comparison were taken from the onboard state estimates provided by the PX4 flight controller through its state-estimation framework, rather than from a single raw sensor channel. These exported state-estimation logs were sampled at 10 Hz, so much of the higher-frequency vibration and measurement noise had already been attenuated before the comparison was performed. The resulting simulated pitch response was then compared with the corresponding flight-test data recorded in the onboard logs. The flight-test data used for comparison were taken from the onboard flight-control logs and therefore contain practical measurement and estimation errors. Accordingly, the agreement between the simulation and flight data should be interpreted as engineering-level consistency rather than exact one-to-one agreement. As illustrated in Figure 6, the simulation-predicted pitch attitude (blue lines) shows a high degree of agreement with the real-world flight data. Over the cruise-validation interval, the comparison between the simulated and flight-test pitch attitudes yields an RMSE of 1.47 deg, an MAE of 1.21 deg, and a maximum absolute error of 3.34 deg, where RMSE and MAE represent the overall error level and the mean absolute deviation, respectively.

The close agreement between the simulation and flight results indicates that the proposed modeling framework captures the main aerodynamic and physical characteristics of the actual vehicle. Specifically, this validates the accuracy of the OpenVSP-based aerodynamic database and the vector thrust coupling formulation. Consequently, this high-fidelity model serves as a reliable digital baseline for the subsequent hybrid optimization process.

5.2. Optimization Result Analysis

The proposed two-stage hybrid optimization strategy was applied to tune the controller parameters for the cruise phase. The step-response analysis in this section was conducted under a nominal cruise condition, where the UAV initially maintained stable cruise flight at 25 m/s with a pitch angle of 5°, and the commanded pitch angle was then changed to 15°. This section evaluates the optimization efficacy by first analyzing the step response characteristics and quantitative metrics, followed by a verification of trajectory tracking performance under complex flight commands. The optimization process is further summarized as follows. Starting from the initial gain set $K_0 = [2.0, 0.35, 0.2, 0, 0.15]$, the Pattern Search stage reduced the objective function value from 1.282×10^7 to 0.19905 in 51 iterations, yielding an intermediate solution $K_{ps} = [5.2500, 0.7094, 0.8875, 0, 0.0250]$. Using this result as the initial guess, the subsequent SQP stage further refined the gains to

$K_{opt} = [5.3149, 0.5215, 0.9677, 0.0143, 0.0389]$, and reduced the objective function to 0.19549 in 12 iterations. These results indicate that the first stage mainly provides coarse search and feasible-solution localization, whereas the second stage performs local refinement of the controller gains.

The dynamic performance improvement is visually demonstrated in Figure 7, where the step response characteristics of the system with baseline and optimized parameters are compared. The baseline response exhibits a sluggish rising edge and significant delay. In contrast, the system with optimized parameters achieves a rapid ascent that eliminates the startup delay observed in the baseline result. These visual observations are confirmed by the quantitative metrics detailed in Table 4. The results show that the optimization process reduced the rise time by approximately 97%, from 6.8 s to 0.2 s. The settling time also decreased significantly, from 10.1 s to 0.9 s, while the steady-state error was reduced to 0.0001 deg. As shown in Table 4 and Figure 7, the optimized response exhibits a small overshoot, whereas the baseline response does not. This is because the baseline response is slower, while the optimized response accepts a limited overshoot within the prescribed constraint range in exchange for faster rise and settling responses. These substantial performance gains indicate that the proposed two-stage optimization procedure identified an improved feasible parameter set within the predefined constraints. The corresponding actuator command remains well within the implemented PWM limit of ± 500 , with a maximum absolute value of 258.18 in the optimized case. This indicates that the integrated vector-thrust longitudinal control mechanism of the ETS30 has sufficient control authority to produce a substantially faster pitch response while still retaining a clear command margin, rather than forcing the actuator to operate at its physical limit. Consequently, the optimized parameter set ensures that the aircraft response is simultaneously rapid, stable, and accurate.

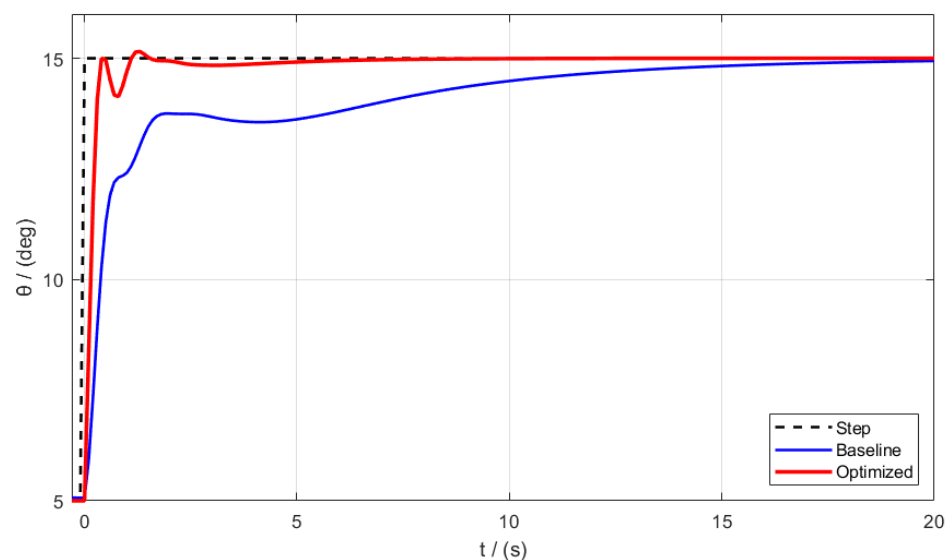


Figure 7. Step response characteristics comparison between baseline and optimized parameter sets.

Table 4. Comparison of dynamic response metrics.

Pitch	Baseline	Optimized
Pitch PP	2.0	5.3149
Pitch rate [KP, KI, KD]	[0.35, 0.2, 0.0]	[0.5215, 0.9677, 0.0143]
Feedforward FF	0.15	0.0389
t_r , s	6.8	0.2
t_s , s	10.1	0.9

Overshoot	0	1.526%
Steady-state error, deg	0.0602	0.0001

To provide a preliminary robustness assessment, the optimized controller was further evaluated under moderate CG perturbations while keeping the controller gains unchanged. Figure 8 shows the corresponding step responses for the optimized nominal case and the perturbed cases with $\pm 10\%$ CG variation. The results indicate that the optimized closed-loop system remains stable under these perturbations, although some degradation in transient performance is observed. For the $+10\%$ CG case, the rise time remains 0.2 s, the settling time increases slightly from 0.9 s to 1.0 s, and the overshoot increases from 1.526% to 6.309%. For the -10% CG case, the rise time becomes 0.3 s, the settling time becomes 0.8 s, and the overshoot becomes 3.571%, while the steady-state error remains close to zero in all cases. These results suggest that the optimized controller is not limited to a single nominal operating point, but retains acceptable pitch-response behavior under moderate longitudinal configuration uncertainty.

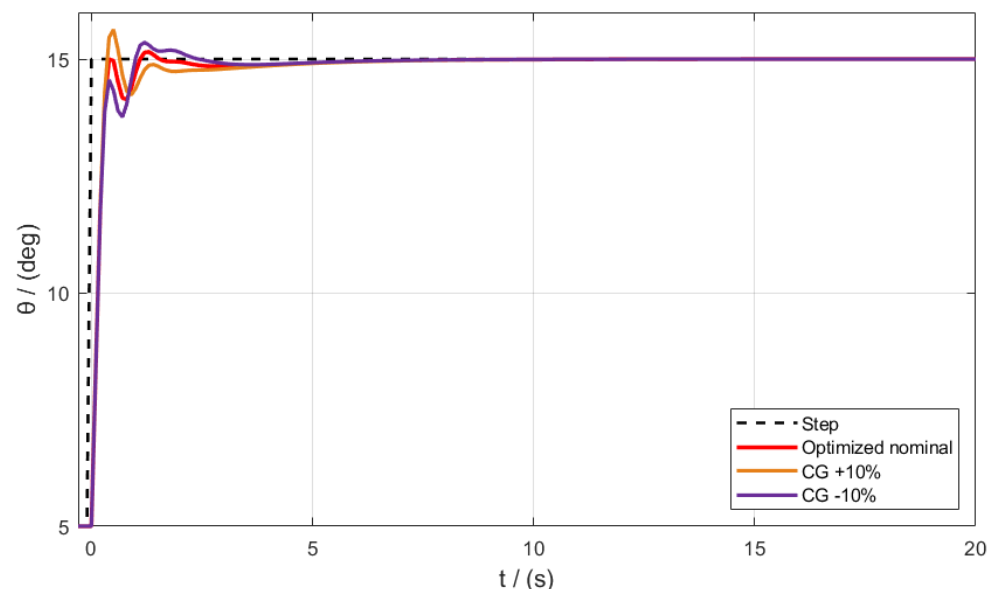


Figure 8. Pitch step responses of the optimized closed-loop system under nominal and perturbed CG conditions with fixed controller gains.

The dynamic tracking performance of the optimized system is further verified through the trajectory tracking test shown in Figure 9, where the actual flight control signals serve as the reference command. The simulation using baseline parameters suffers from significant amplitude attenuation and phase lag when attempting to follow the command input. In contrast, the simulation using the optimized parameters follows the flight-command curve more closely than the baseline case. The improved tracking capability confirms that the closed-loop bandwidth has been effectively expanded by the optimization process. It is worth noting that the minor phase lag persisting in the optimized response is an intrinsic physical characteristic attributed to the necessary aerodynamic damping.

The present workflow combines an OpenVSP-based aerodynamic database, a cruise-phase nonlinear longitudinal model, and a two-stage controller-parameter optimization strategy using Pattern Search and SQP. Within this framework, the aerodynamic database provides the force and moment inputs for simulation, the nonlinear model supports the cruise-response analysis, and the two-stage optimization process enables coarse exploration followed by local refinement under prescribed dynamic constraints. Although the present study focuses on a vector-thrust tail-sitter UAV in cruise flight, the same workflow

can be extended to other VTOL UAVs or fixed-wing configurations after redefining the corresponding aerodynamic/propulsion database and control constraints.

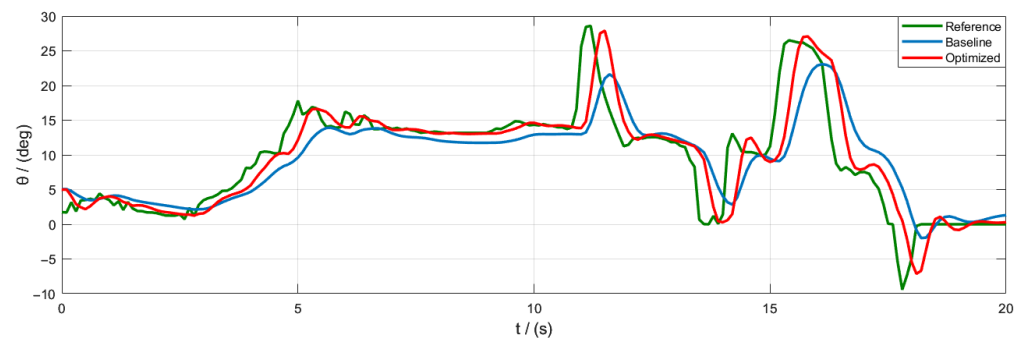


Figure 9. Trajectory tracking performance comparison between baseline and optimized controllers under flight command inputs.

6. Conclusions

This paper focuses on the cruise-phase longitudinal modeling and control-parameter optimization of the ETS30 tail-sitter UAV. A nonlinear dynamic model characterizing the coupling characteristics between longitudinal aerodynamics and vector propulsion was established. The validity of the model was verified through flight experiments, which confirmed a high consistency between the simulation and flight test. Based on the verified model, a two-stage hybrid strategy was utilized for control parameter optimization, identifying an improved feasible parameter set within the prescribed constraints. The results demonstrate that the optimized control parameters significantly enhance pitch stability and dynamic response speed. Specifically, the rise time was reduced by approximately 97% compared to the baseline, ensuring rapid and precise trajectory tracking performance. The present framework is applicable to the cruise longitudinal channel of the platform, rather than to the full flight envelope, including hover and transition. Future work will extend the present framework to hover and transition conditions, where additional multi-rotor coupling and mode-switching dynamics must be considered.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/drones10050316/s1>, Video S1: ETS30 flight test.

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Abbreviations

The following abbreviations are used in this manuscript:

CG	Center of gravity
PID	Proportional integral derivative
UAV	Unmanned aerial vehicle
SQP	Sequential Quadratic Programming
VTOL	Vertical Take-Off and Landing

Appendix A

Table A1 lists the nomenclature used in this paper.

Table A1. Nomenclature.

Symbol	Physical Significance
c	chord length
Q	dynamic pressure
S	wing area
l_t	the distance from the tail motor to the center of gravity
m	mass of ETS30
q	pitch rate
C_L	lift coefficient
C_D	drag coefficient
C_m	total pitching moment coefficient
I_{yy}	inertia moment about the Y-axis
L	lift
V	velocity
α	angle of attack
θ	pitch angle
γ	flight path angle
h	the height of ETS30
t_p	peak time
t_r	rise time
t_s	settling time
M_p	the maximum overshoot
e_{ss}	steady state error
δ_{vec}	the deflection angle of the tail motor relative to the body axis

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